Computer Networks: Complete Notes

Unit 1: Introduction

1.1 What is a computer network?

A set of interconnected devices (hosts, routers, switches) that exchange data over a medium using agreed rules called **protocols**.

Uses: resource sharing, communication (email, VoIP), remote access, cloud services, e-commerce.

1.2 Network types by size

Type	Area	Example
PAN	A few meters	Bluetooth headset
LAN	Building / campus	Office Wi-Fi, Ethernet
MAN	City	Cable TV network, city Wi-Fi
WAN	Country / world	The Internet

1.3 Topologies

Topology	Description	Pros	Cons
Bus	One shared cable	Cheap, simple	Single point of failure, collisions
Ring	Each node connects to two neighbors	Orderly access	One break can stop the ring
Star	All nodes connect to a central hub/switch	Easy to manage, one node failing is isolated	Central device is a single point of failure
Mesh	Every node connects to every other	Very reliable	Expensive. Links = n(n-1)/2
Tree	Hierarchy of stars	Scalable	Root dependency
Hybrid	Combination	Flexible	Complex

1.4 Transmission modes

- **Simplex:** one direction only (keyboard to computer).
- **Half duplex:** both directions, one at a time (walkie-talkie).
- **Full duplex:** both directions at once (telephone).

1.5 Switching techniques

- **Circuit switching:** a dedicated path is reserved for the whole session (old telephone network). Guaranteed bandwidth, but wasteful.
- **Packet switching:** data is split into packets, each routed independently (the Internet). Efficient sharing, but variable delay.
- **Datagram:** each packet routed separately, can arrive out of order.
- **Virtual circuit:** a path is set up first, packets follow it.

- **Message switching:** whole message stored and forwarded at each node.

1.6 Delays in packet networks

Total nodal delay = processing + queuing + transmission + propagation.

- **Transmission delay** = L / R (packet length / link rate)
- **Propagation delay** = d / s (distance / propagation speed)
- **Round-trip time (RTT)** = time for a packet to go and the reply to return.

Example: 1000-byte packet on a 1 Mbps link: transmission delay = 8000 bits / 10^6 = 8 ms.

1.7 Performance metrics

- **Bandwidth:** capacity of the link (bits/s).
- **Throughput:** actual achieved rate.
- **Latency:** time for data to travel.
- **Jitter:** variation in latency.
- **Bandwidth-delay product** = bandwidth x RTT: the amount of data "in flight" on the link.

Unit 2: Network Models

2.1 OSI model (7 layers)

#	Layer	Function	PDU	Examples
7	Application	Interface for network services	Data	HTTP, FTP, SMTP, DNS
6	Presentation	Translation, encryption, compression	Data	SSL/TLS, JPEG, ASCII
5	Session	Establish, manage, terminate sessions	Data	NetBIOS, RPC
4	Transport	End-to-end delivery, reliability, flow control	Segment / Datagram	TCP, UDP
3	Network	Logical addressing, routing	Packet	IP, ICMP, OSPF
2	Data Link	Framing, MAC addressing, error detection	Frame	Ethernet, PPP, Wi-Fi
1	Physical	Bits on the medium	Bits	Cables, hubs, signals

Memory aid (top to bottom): **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing.

2.2 TCP/IP model (4 layers)

TCP/IP layer	OSI equivalent
Application	Application, Presentation, Session
Transport	Transport

TCP/IP layer	OSI equivalent
Internet	Network
Network Access (Link)	Data Link, Physical

2.3 Encapsulation

Each layer adds its own header (and sometimes trailer) as data moves **down**, and removes it as data moves **up** at the receiver.

```
Application data
  + TCP header         -> Segment
  + IP header          -> Packet
  + Ethernet header/trailer -> Frame
  -> bits on the wire
```

Unit 3: Physical Layer

3.1 Transmission media

Guided: twisted pair (UTP/STP), coaxial cable, optical fiber.

Unguided: radio waves, microwaves, infrared, satellite.

Medium	Speed / Range	Notes
UTP (Cat5e/6)	Up to 1-10 Gbps, ~100 m	Cheap, common in LANs
Coaxial	Higher bandwidth than UTP	Cable TV
Fiber	Very high, long distance	Immune to EMI, expensive

3.2 Signals and encoding

- **Analog** (continuous) vs **digital** (discrete).
- **Line coding:** NRZ, NRZ-I, Manchester (used in classic Ethernet; clock and data combined), Differential Manchester.
- **Modulation:** ASK, FSK, PSK, QAM.

3.3 Channel capacity formulas

- **Nyquist (noiseless):** Max bit rate = 2 x B x log■(L), where B is bandwidth in Hz and L is number of signal levels.
- **Shannon (noisy):** Capacity = B x log■(1 + SNR).

Example: B = 3000 Hz, SNR = 1023. Capacity = 3000 x log■(1024) = 30,000 bps.

3.4 Multiplexing

Sharing one link among multiple signals.

- **FDM:** different frequency bands.
- **TDM:** different time slots (synchronous or statistical).
- **WDM:** different light wavelengths on fiber.

3.5 Physical devices

- **Repeater:** regenerates signals.
- **Hub:** multi-port repeater; one collision domain.

Unit 4: Data Link Layer

4.1 Functions

Framing, physical addressing (MAC), error control, flow control, access control.

4.2 Framing methods

Character count, flag bytes with **byte stuffing**, flag bits with **bit stuffing** (insert 0 after five consecutive 1s when the flag is 01111110), physical-layer coding violations.

4.3 Error detection and correction

Parity bit: adds 1 bit so the total number of 1s is even (or odd). Detects single-bit errors only.

Checksum: sum of data words, complemented. Used by IP, TCP and UDP.

CRC (Cyclic Redundancy Check):

1. Append (n) zeros to the data, where n = degree of generator polynomial.
2. Divide by the generator using **modulo-2 division** (XOR).
3. The remainder is the CRC, appended to the data.
4. Receiver divides the whole frame. Remainder 0 means no detected error.

Hamming code: corrects single-bit errors. Parity bits at positions 1, 2, 4, 8, ... (powers of 2). Need 2^r >= m + r + 1, where m = data bits, r = parity bits.

Hamming distance: to detect d errors need distance d + 1; to correct d errors need 2d + 1.

4.4 Flow control protocols (ARQ)

Let W = window size, RTT-related terms as below.

Protocol	Sender window	Receiver window	Behavior on error
Stop-and-Wait	1	1	Retransmit after timeout
Go-Back-N	N (up to 2^k - 1)	1	Retransmit the lost frame and all after it
Selective Repeat	N (up to 2^(k-1))	N	Retransmit only the lost frame

Efficiency (Stop-and-Wait) = 1 / (1 + 2a), where a = propagation delay / transmission delay.

Sliding window efficiency = min(1, N / (1 + 2a)).

For k-bit sequence numbers: Go-Back-N sender window max = 2^k - 1; Selective Repeat window max = 2^(k-1).

4.5 Medium Access Control (MAC)

Protocol	Idea
Pure ALOHA	Send anytime; collisions are retried. Max efficiency 18.4% (1/2e)
Slotted ALOHA	Send only at slot starts. Max efficiency 36.8% (1/e)
CSMA	Listen before sending
CSMA/CD	Listen while sending; stop on collision (Ethernet). Needs minimum frame size so the sender detects collision: **Frame time >= 2 x propagation delay**
CSMA/CA	Collision **avoidance** with RTS/CTS and backoff (Wi-Fi 802.11)
Token Ring	A token circulates; only the holder transmits

Binary exponential backoff (Ethernet): after the i-th collision, wait a random number of slots in [0, 2^i - 1].

4.6 Ethernet (IEEE 802.3)

Frame format: Preamble (7 B) | SFD (1 B) | Dest MAC (6 B) | Src MAC (6 B) | Type/Length (2 B) | Data (46-1500 B) | FCS (4 B).

- **Minimum frame:** 64 bytes. **Maximum payload (MTU):** 1500 bytes.
- **MAC address:** 48 bits, written as six hex pairs, e.g., `00:1A:2B:3C:4D:5E`. First 24 bits identify the manufacturer (OUI).

4.7 Switches, bridges, VLANs

- **Bridge / switch:** forwards frames using a **MAC address table**, learned from source addresses. Each port is a separate collision domain.
- **Switch behavior:** if destination is known, forward to that port; if unknown, **flood**; if on the same port, **filter**.
- **VLAN:** logical segmentation of a LAN into separate broadcast domains.
- **Spanning Tree Protocol (STP):** prevents loops in switched networks.

4.8 ARP

Maps an **IP address to a MAC address** on the local network. The host broadcasts "Who has IP X?"; the owner replies with its MAC. Results are cached.

Unit 5: Network Layer

5.1 Functions

Logical addressing, routing, forwarding, fragmentation, congestion control.

5.2 IPv4 addressing

32-bit address in dotted decimal (e.g., `192.168.1.10`).

Classful addressing

Class	First octet	Default mask	Hosts per network
A	1-126	255.0.0.0 (/8)	2^24 - 2
B	128-191	255.255.0.0 (/16)	2^16 - 2
C	192-223	255.255.255.0 (/24)	2^8 - 2 = 254
D	224-239	Multicast	
E	240-255	Reserved	

(127.x.x.x is loopback.)

Private ranges (RFC 1918): 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.

5.3 Subnetting and CIDR

CIDR notation: `a.b.c.d/n`, where n is the number of network bits.

- Hosts per subnet = 2^(32 - n) - 2 (subtract network and broadcast addresses).
- Number of subnets from borrowing b bits = 2^b.

Example: `192.168.10.0/26`

- Mask: 255.255.255.192
- Block size: 64
- Hosts per subnet: 2^6 - 2 = 62
- Subnets: 0, 64, 128, 192
- For subnet 192.168.10.64: network address .64, usable .65 to .126, broadcast .127.

VLSM: different subnet sizes inside one network, allocating the largest requirement first.

Supernetting / route aggregation: combine contiguous networks into one prefix, e.g., 192.168.0.0/24 and 192.168.1.0/24 become 192.168.0.0/23.

5.4 IPv4 header (20 bytes minimum)

Version | IHL | ToS | Total Length | Identification | Flags (DF, MF) | Fragment Offset | **TTL** | Protocol | Header Checksum | Source IP | Destination IP | Options.

- **TTL** decreases by 1 at each router; at 0 the packet is discarded (prevents loops).
- **Fragmentation:** if packet > MTU, the router splits it. Fragment offset is in units of 8 bytes. Reassembly happens only at the destination.

5.5 IPv6

128-bit addresses (e.g., `2001:db8::1`), simplified 40-byte fixed header, no fragmentation by routers, no header checksum, built-in IPSec support, no ARP (uses Neighbor Discovery).

Transition: dual stack, tunneling, translation (NAT64).

5.6 Supporting protocols

- **ICMP:** error and diagnostic messages (`ping`, `traceroute`). Examples: destination unreachable, time exceeded.
- **DHCP:** automatically assigns IP, mask, gateway, DNS. Process: **DORA** (Discover, Offer, Request, Acknowledge).

- **NAT:** translates private IPs to a public IP (and ports) so many hosts share one address.
- **IGMP:** manages multicast group membership.

5.7 Routing

Static vs dynamic routing. Dynamic routers learn routes from protocols.

Distance Vector (RIP):

- Routers share their whole table with **neighbors** periodically.
- Uses **Bellman-Ford**. Metric = hop count (max 15; 16 = infinity).
- Problem: **count-to-infinity**. Fixes: split horizon, poison reverse, hold-down timers.

Link State (OSPF):

- Each router floods **link-state advertisements** to all routers, builds the full topology map, runs **Dijkstra's algorithm**.
- Faster convergence, more memory and CPU. Uses areas to scale.

BGP: the path-vector protocol that exchanges routes **between autonomous systems** on the Internet (inter-domain). RIP and OSPF are intra-domain (IGPs).

	Distance Vector	Link State
Knowledge	Neighbors' info	Whole topology
Algorithm	Bellman-Ford	Dijkstra
Updates	Periodic, to neighbors	Event-triggered, flooded
Convergence	Slow	Fast

5.8 Congestion control (network layer view)

Traffic shaping with **leaky bucket** (constant output rate) and **token bucket** (allows bursts); admission control; load shedding; choke packets.

Unit 6: Transport Layer

6.1 Functions

Process-to-process delivery using **port numbers**, segmentation, multiplexing/demultiplexing, reliability, flow and congestion control.

A **socket** = IP address + port number. A connection is identified by (src IP, src port, dest IP, dest port).

Port ranges: well-known 0-1023, registered 1024-49151, dynamic 49152-65535.

6.2 UDP

Connectionless, unreliable, no ordering, no congestion control. Header is only 8 bytes (src port, dest port, length, checksum).

Uses: DNS, DHCP, streaming, VoIP, online gaming.

6.3 TCP

Connection-oriented, reliable, ordered, byte-stream, full duplex. Header 20 bytes minimum.

Header fields: Source port, Dest port, Sequence number, Acknowledgment number, Header length, Flags (URG, ACK, PSH, RST, SYN, FIN), Window size, Checksum, Urgent pointer, Options.

Three-way handshake (connection setup)

1. Client sends `SYN` (seq = x).
2. Server replies `SYN` + `ACK` (seq = y, ack = x + 1).
3. Client sends `ACK` (ack = y + 1).

Connection termination (four-way): `FIN`, `ACK`, `FIN`, `ACK`. The side that closes first enters **TIME_WAIT**.

Reliability: sequence numbers, ACKs, retransmission on timeout or on **3 duplicate ACKs** (fast retransmit).

Timeout estimation:

- EstimatedRTT = (1 - α) x EstimatedRTT + α x SampleRTT (α = 0.125)
- DevRTT = (1 - β) x DevRTT + β x |SampleRTT - EstimatedRTT| (β = 0.25)
- Timeout = EstimatedRTT + 4 x DevRTT

6.4 TCP flow control

The receiver advertises a **receive window (rwnd)** so the sender never overruns its buffer. Sender can have at most min(cwnd, rwnd) unacknowledged bytes.

6.5 TCP congestion control

- **Slow start:** cwnd starts at 1 MSS and **doubles every RTT** (exponential) until reaching **ssthresh**.
- **Congestion avoidance:** cwnd grows by 1 MSS per RTT (additive increase).
- **On timeout:** ssthresh = cwnd / 2; cwnd = 1 MSS; go back to slow start (TCP Tahoe/Reno).
- **On 3 duplicate ACKs (Reno):** ssthresh = cwnd / 2; cwnd = ssthresh (+3); fast recovery.
- Overall behavior is **AIMD** (additive increase, multiplicative decrease).

6.6 TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed	Best effort
Ordering	Yes	No
Speed / overhead	Slower, heavier	Faster, lighter
Header	20+ bytes	8 bytes
Use cases	Web, email, file transfer	DNS, streaming, gaming

Unit 7: Application Layer

7.1 Client-server vs peer-to-peer

Client-server: a server with a fixed address serves many clients (web). P2P: peers act as both client and server (BitTorrent).

7.2 DNS (Domain Name System)

Translates names (e.g., `www.example.com`) to IP addresses. Hierarchical, distributed database.

- **Hierarchy:** Root, TLD (.com, .org, .in), authoritative servers.
- **Resolution:** Recursive (resolver does all the work) or iterative (server returns the next server to ask). Results are **cached** with a TTL.
- **Record types: A** (name to IPv4), **AAAA** (name to IPv6), **CNAME** (alias), **MX** (mail server), **NS** (name server), **PTR** (reverse lookup), **TXT**.
- Uses UDP port 53 (TCP for large responses and zone transfers).

7.3 HTTP

Request-response protocol on TCP port 80 (HTTPS: 443). Stateless.

- **Methods:** GET (read), POST (create/submit), PUT (replace), PATCH (partial update), DELETE, HEAD, OPTIONS.
- **Status codes:** 1xx informational, 2xx success (200 OK, 201 Created), 3xx redirect (301, 302, 304 Not Modified), 4xx client error (400, 401, 403, 404), 5xx server error (500, 502, 503).
- **HTTP/1.0** one request per connection; **HTTP/1.1** persistent connections, pipelining; **HTTP/2** multiplexing over one connection, header compression; **HTTP/3** runs over QUIC (UDP).
- **Cookies** add state; **caching** and **proxies** reduce load.

Sample request

```
GET /index.html HTTP/1.1
Host: www.example.com
Accept: text/html
```

7.4 Email

- **SMTP** (port 25/587): sending mail between servers (push).
- **POP3** (110): download mail and typically delete from the server.
- **IMAP** (143): manage mail on the server, multiple devices stay in sync.
- **MIME:** lets email carry attachments and non-ASCII text.

7.5 Other protocols

Protocol	Port	Purpose
FTP	20 / 21	File transfer (data / control)
SFTP / SSH	22	Secure file transfer, remote login
Telnet	23	Remote login (insecure)
SMTP	25	Send email
DNS	53	Name resolution
DHCP	67 / 68	IP assignment
HTTP	80	Web
POP3	110	Receive email

Protocol	Port	Purpose
IMAP	143	Receive email
HTTPS	443	Secure web

Unit 8: Network Security

8.1 Goals (CIA triad)

Confidentiality, **Integrity**, **Availability**. Also authentication and non-repudiation.

8.2 Cryptography

- **Symmetric** (same key to encrypt and decrypt): AES, DES. Fast; key distribution is the challenge.
- **Asymmetric** (public/private key pair): RSA, ECC. Slower; solves key exchange and enables digital signatures.
- **Hash functions** (SHA-256): fixed-size, one-way digest for integrity.
- **Digital signature:** hash of message encrypted with the sender's **private key**; anyone can verify with the public key.
- **Digital certificate:** binds a public key to an identity, signed by a Certificate Authority (CA).
- **Diffie-Hellman:** two parties agree on a shared secret over an insecure channel.

8.3 Secure protocols

- **TLS/SSL:** encrypts data between client and server (HTTPS). Handshake: negotiate cipher, authenticate server via certificate, exchange keys, then use symmetric encryption for the session.
- **IPSec:** security at the network layer (modes: transport, tunnel; protocols: AH, ESP). Basis of many VPNs.
- **SSH:** secure remote shell.
- **WPA2/WPA3:** Wi-Fi security.

8.4 Common attacks and defenses

Attack	Description	Defense
Sniffing	Capturing traffic	Encryption
Spoofing	Faking IP/MAC	Ingress filtering, authentication
Man-in-the-middle	Intercepting and altering traffic	TLS, certificates
DoS / DDoS	Overwhelming a service	Rate limiting, CDN, filtering
SYN flood	Many half-open TCP connections	SYN cookies
DNS spoofing	Fake DNS responses	DNSSEC
Phishing	Tricking users	Awareness, filtering
ARP poisoning	False ARP replies	Dynamic ARP inspection

8.5 Network security devices

- **Firewall:** filters traffic by rules (packet filter, stateful, application-level/proxy).
- **IDS / IPS:** detect / prevent malicious activity.
- **VPN:** encrypted tunnel across a public network.

- **DMZ:** isolated zone for public-facing servers.

Unit 9: Wireless and Modern Networks

- **Wi-Fi (IEEE 802.11):** uses CSMA/CA. Standards: 802.11b/g/n/ac/ax (Wi-Fi 6). Modes: infrastructure (via access point) and ad hoc. Problems: **hidden terminal** and **exposed terminal**, solved partly with RTS/CTS.
- **Bluetooth (802.15.1):** short-range PAN.
- **Cellular:** 2G (GSM), 3G (UMTS), 4G (LTE), 5G (high speed, low latency, network slicing).
- **SDN (Software Defined Networking):** separates the **control plane** from the **data plane**; a central controller programs the switches (e.g., OpenFlow).
- **CDN:** servers distributed worldwide to deliver content closer to users.
- **IoT:** large numbers of low-power devices (MQTT, CoAP, 6LoWPAN).

Quick Reference: Important Formulas

Topic	Formula
Transmission delay	L / R
Propagation delay	d / s
Nyquist	2 B log■ L
Shannon	B log■(1 + SNR)
Stop-and-wait efficiency	1 / (1 + 2a), a = Tp / Tt
Sliding window efficiency	min(1, N / (1 + 2a))
GBN max window	2^k - 1
SR max window	2^(k-1)
Pure ALOHA max efficiency	18.4%
Slotted ALOHA max efficiency	36.8%
Hosts in a subnet	2^(32 - n) - 2
Bandwidth-delay product	Bandwidth x RTT
CSMA/CD min frame time	>= 2 x Tp
TCP timeout	EstRTT + 4 x DevRTT

Practice Questions

1. Differentiate between OSI and TCP/IP models.
2. A 1500-byte packet is sent over a 10 Mbps link, 2000 km long, with propagation speed 2 x 10^8 m/s. Find transmission delay, propagation delay and total delay.
3. Divide 192.168.5.0/24 into 4 equal subnets. Write network, first host, last host and broadcast addresses.
4. Explain TCP three-way handshake and why two-way is not enough.

5. Compare Go-Back-N and Selective Repeat.
6. Explain distance vector routing and the count-to-infinity problem.
7. What happens when you type `www.example.com` in a browser and press Enter? (DNS, TCP, TLS, HTTP.)
8. Compare TCP and UDP, and give two applications for each.
9. Explain how a switch learns MAC addresses.
10. Explain TCP slow start and congestion avoidance with a cwnd graph.

Answers (selected)

Q2. Transmission delay = 12000 bits / 10^7 = 1.2 ms. Propagation = 2×10^6 m / 2×10^8 m/s = 10 ms. Total = 11.2 ms (ignoring processing and queuing).

Q3. Borrow 2 bits, so /26, block size 64.

Subnet	Network	First host	Last host	Broadcast
1	192.168.5.0	.1	.62	.63
2	192.168.5.64	.65	.126	.127
3	192.168.5.128	.129	.190	.191
4	192.168.5.192	.193	.254	.255

Q4. The three-way handshake lets both sides confirm that the other can send and receive and synchronize initial sequence numbers. With only two steps, the server cannot know the client received its SYN, and old duplicate SYNs could create false connections.

Q7. Browser checks cache, then DNS resolves the name to an IP (possibly via root, TLD, authoritative servers). A TCP handshake opens a connection to port 443. TLS handshake authenticates the server and sets up encryption. The browser sends an HTTP GET request, receives the response, and renders the page, fetching other resources as needed.

End of notes.